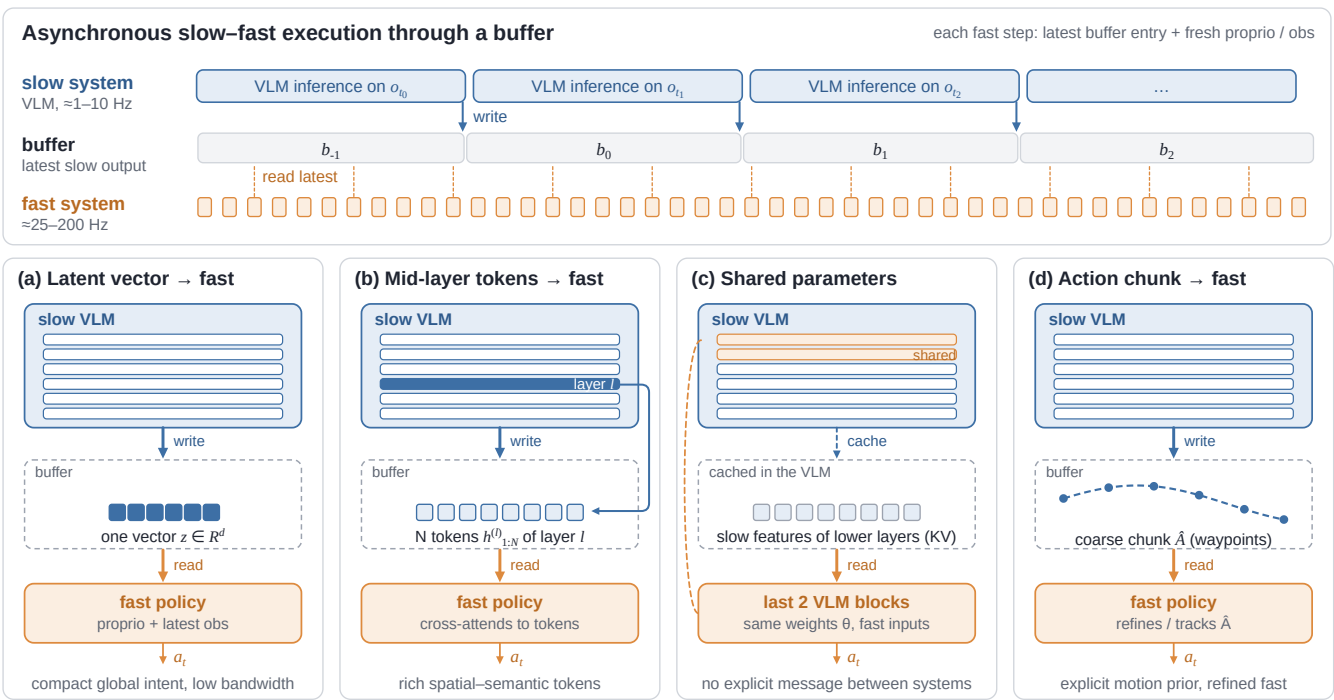

Asynchronous slow–fast execution through a buffer
each fast step: latest buffer entry + fresh proprio / obs
slow system
VLM, ≈1–10 Hz
buffer
latest slow output
fast system
≈25–200 Hz
VLM inference on o_{t_0}
VLM inference on o_{t_1}
VLM inference on o_{t_2}
…
write
b_{-1}
b_0
b_1
b_2
read latest
(a) Latent vector → fast
slow VLM
write
buffer
one vector $z \in R^d$
read
fast policy
proprio + latest obs
a_t
compact global intent, low bandwidth
(b) Mid-layer tokens → fast
slow VLM
layer l
write
buffer
N tokens $h^{(l)}_{1:N}$ of layer l
read
fast policy
cross-attends to tokens
a_t
rich spatial–semantic tokens
(c) Shared parameters
slow VLM
shared
cache
cached in the VLM
slow features of lower layers (KV)
read
last 2 VLM blocks
same weights θ, fast inputs
a_t
no explicit message between systems
(d) Action chunk → fast
slow VLM
write
buffer
coarse chunk $\hat{A}$ (waypoints)
read
fast policy
refines / tracks $\hat{A}$
a_t
explicit motion prior, refined fast